A

INTERNSHIP REPORT ON

"Full Stack Web Application using MERN Stack"

SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY IN THE PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE

OF

THIRD YEAR OF ENGINEERING

(COMPUTER ENGINEERING)

SUBMITTED BY

Mr./Ms.:- PAWAR SIDDHESH SURYAKANT

Exam Seat No. T400430419

Area of Internship:- Full Stack Web Application using MERN Stack

Sem:- VI

Sinhgad Institutes

DEPARTMENT OF COMPUTER ENGINEERING

STES'S SINHGAD ACADEMY OF ENGINEERING

KONDHWA, PUNE 411048

UNIVERSITY OF PUNE

2025-26

Sinhgad Institutes

DEPARTMENT OF COMPUTER ENGINEERING

STES'S SINHGAD ACADEMY OF ENGINEERING
KONDHWA, PUNE 411048

CERTIFICATE

This is to certify that the Internship report entitle

"Full Stack Web Application using MERN Stack"

Submitted by

Mr./Ms. PAWAR SIDDHESH SURYAKANT Exam No: T400430419

is a bonafide work carried out by him/her under the supervision of **Prof. A.H. Kagane** and it is submitted towards the Partial fulfillment of the requirement of Savitribai Phule Pune University, for the award of the degree of Third year of Engineering (Computer Engineering).

Prof .A.H. Kagane
TG/Mentor
Department of Computer Engineering

Prof. S. N. Shelke
Head,
Department of Computer Engineering

Dr. K.P.Patil
Principal,
Sinhgad Academy of Engineering, Pune – 48

Place : Pune

Date :

ACKNOWLEDGEMENT

It is indeed with a great sense of pleasure and immense sense of gratitude that I acknowledge the help of these individuals

Working in the **(CODEC TECHNOLOGIES)** was interesting. During these one months of internship, I learnt a lot web development, especially on completing three task all are helpful for understanding all concepts in HTML, CSS and JavaScript.

I have to thank **(CODEC TECHNOLOGIES))**for giving opportunity and platform for internship.

Therefore, I am grateful to the people in the **(CODEC TECHNOLOGIES))**for the chance to make this experiment. And Opportunity to build a three small variant task and that very helpful to my knowledge.

Further on, I want to thank the students and interns in the **(CODEC TECHNOLOGIES))**who made this demanding time joyful but always efficient.

I am extremely great full to my department Internship Coordinator **Prof. A.H.Kagane** and H.O.D. Prof. **S.N.Shelke** Sir who Guided and helped me in successful completion of this internship.

PAWAR SIDDHESH SURYAKANT

Seat No. T400430419

ABSTRACT

This report presents the details of my internship training in MERN Stack Development at CODEC Technologies. The MERN stack consists of MongoDB, Express.js, React.js, and Node.js, which together form a powerful framework for building dynamic web applications.

During this training, I gained knowledge of both frontend and backend technologies. I learned how to design responsive user interfaces using React and how to manage server-side operations using Node.js and Express.js. Additionally, I worked with MongoDB for database management.

The training also included practical sessions where I developed real-world projects. These projects helped me understand how to integrate frontend and backend components effectively. I also learned about RESTful APIs and how to handle HTTP requests.

Version control tools like Git were introduced, enabling efficient code management. The internship emphasized hands-on learning, problem-solving, and debugging techniques. Overall, this training enhanced my technical skills and prepared me for real-world software development challenges. It provided a strong foundation in full-stack development and improved my ability to build scalable web applications.

CONTENTS

Sr.No	TITLE	Page no
1.	Acknowledgement	
2.	Abstract	
3.	Internship Offer and Completion Certificate(Two Scanned Copies of each)	
4.	Internship Place Details: Company Background..	
5.	Introduction	
6.	Title/ Problem Statement	
7.	Objectives	
7.	Motivation/Scope	
8.	Design	
9.	Methodologies Used	
10.	Hardware and Software Used	
11.	Results	
12.	Future Scope	
13.	Conclusion	
14.	References	

INTERNSHIP PLACE DETAILS

The internship was carried out at CODEC Technologies, a reputed institute known for providing high-quality training in the field of information technology and software development. The organization focuses on delivering industry-oriented education and bridging the gap between academic knowledge and practical skills.

CODEC Technologies offers a wide range of courses, including web development, full-stack development, programming languages, data science, and emerging technologies. The institute is committed to equipping students with the latest technical skills required in today's competitive IT industry. Their training programs are designed in alignment with current industry standards and practices.

The organization has a team of experienced trainers who possess strong technical expertise and real-world industry experience. They provide in-depth knowledge through interactive sessions, practical demonstrations, and real-time project guidance. The teaching methodology focuses on "learning by doing," which helps students understand concepts more effectively.

The infrastructure at CODEC Technologies is well-equipped with modern computer systems, high-speed internet connectivity, and updated software tools. The learning environment is professional and encourages students to actively participate in coding activities and discussions.

During the internship, the institute provided structured training in MERN Stack Development, covering MongoDB, Express.js, React.js, and Node.js. The program included theoretical sessions along with hands-on project work, enabling students to gain real-world development experience.

CODEC Technologies also emphasizes soft skills and career development. Students are guided in resume building, interview preparation, and problem-solving techniques. This holistic approach ensures that trainees are not only technically skilled but also professionally prepared.

The organization's mission is to develop skilled professionals who can contribute effectively to the IT industry. Its vision is to become a leading training provider by continuously updating its curriculum and adopting modern teaching techniques.

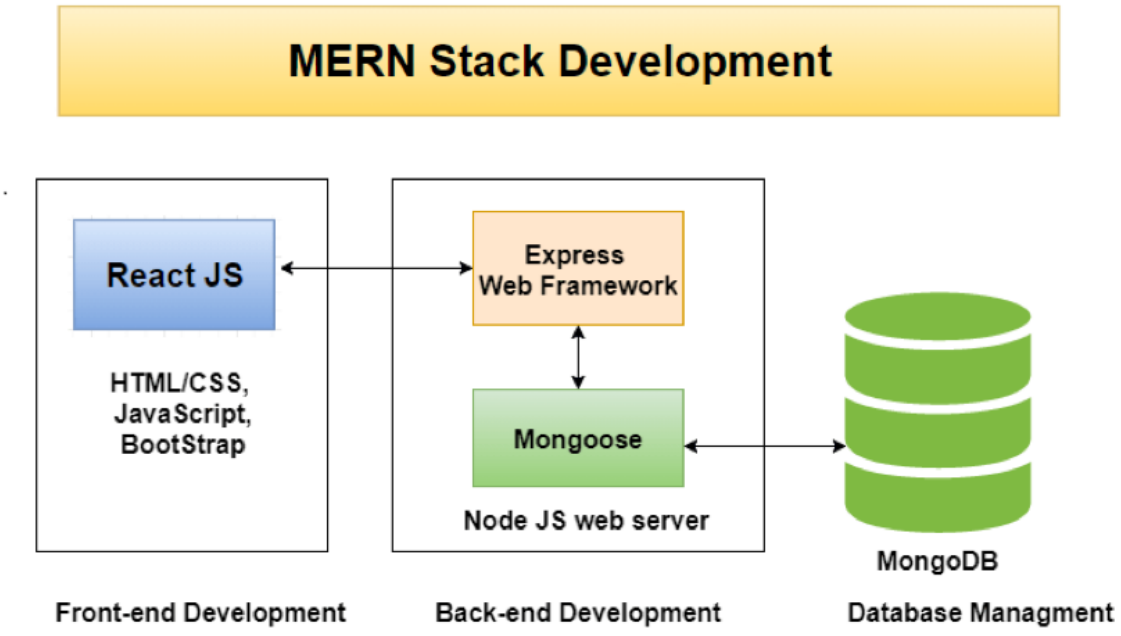
Overall, the internship at CODEC Technologies provided a supportive and motivating environment that enhanced my technical knowledge, practical skills, and confidence in software development.

INTRODUCTION

The MERN stack is a popular technology stack used for developing modern web applications. It includes MongoDB, Express.js, React.js, and Node.js. These technologies work together to provide a complete solution for frontend and backend development.

During my internship at CODEC Technologies, I was introduced to the fundamentals of web development. I learned how different components of a web application interact with each other.

The training started with basic concepts such as HTML, CSS, and JavaScript. Gradually, it progressed to advanced topics like React components, API integration, and database management.
The internship provided practical exposure through project development. This helped me understand how to implement theoretical concepts in real-world scenarios.
The main objective of this training was to build a strong foundation in full-stack development. It also aimed to improve problem-solving and coding skills.
Overall, the introduction to MERN stack has opened new opportunities for me in the field of software development.



TITLE/PROBLEM STATEMENT

The title of my internship project is “Full Stack Web Application using MERN Stack.” The main objective was to develop a dynamic and responsive web application.
The problem statement involved creating a system that can handle user data efficiently and provide a smooth user experience. The application needed to support CRUD operations (Create, Read, Update, Delete).

Another challenge was integrating frontend and backend components seamlessly. Ensuring data security and proper API handling was also important.
The project required implementing authentication and authorization features. This ensures that only authorized users can access certain functionalities.

The system needed to be scalable and efficient. Proper database design was essential to achieve this goal.

Overall, the problem statement focused on building a complete web application using modern technologies.

The problem statement involved creating a system capable of managing user data efficiently while providing a seamless user experience. The application needed to support CRUD operations and ensure data integrity.

One of the major challenges was integrating different technologies into a single application. This required a clear understanding of both frontend and backend processes.
Security was another important aspect, requiring the implementation of authentication and authorization mechanisms.

The application also needed to be scalable and efficient to handle multiple users.
Overall, the project focused on solving real-world problems using modern web technologies.

OBJECTIVES

The main objective of this internship was to learn MERN stack development. It aimed to provide knowledge of both frontend and backend technologies.

- Another objective was to develop practical skills through real-world projects. This helped in understanding how web applications are built.

- The training also focused on improving coding and debugging skills. Learning version control using Git was another important objective.

- Understanding REST APIs and database management was also a key goal. The internship aimed to enhance problem-solving abilities.

- It also focused on building responsive and user-friendly interfaces.
  Overall, the objectives were designed to prepare students for professional roles in software development.
- It aimed to provide hands-on experience in building web applications.

- Another objective was to understand the interaction between frontend and backend systems.

- The training focused on improving coding, debugging, and problem-solving skills.

- Learning database management and API development was also a key goal.

- The internship aimed to prepare students for real-world software development roles.

- It also encouraged teamwork and collaborative learning.

MOTIVATION SCOPE

- The motivation behind learning MERN stack development is the increasing demand for full-stack developers. Companies require professionals who can handle both frontend and backend tasks.

- This internship provided an opportunity to gain such skills. The scope of MERN stack is vast and includes web and mobile application development.

- Learning these technologies opens doors to various career opportunities. It also allows developers to build scalable and efficient applications.

- The training motivated me to explore new technologies and improve my skills.

- The scope also includes working on real-time projects and contributing to open-source platforms.

- Overall, MERN stack development offers great career growth and opportunities.

- The motivation behind learning MERN stack lies in its high demand in the software industry.

- Full-stack developers are highly valued for their ability to handle complete projects.

- This internship provided an opportunity to learn these in-demand skills.

- The scope of MERN stack includes developing web applications, mobile apps, and enterprise solutions.

- It also opens opportunities in freelancing and startup development.

- The training motivated me to continuously learn and upgrade my skills.

- Overall, it provides a strong foundation for a successful career in IT.

DESIGN

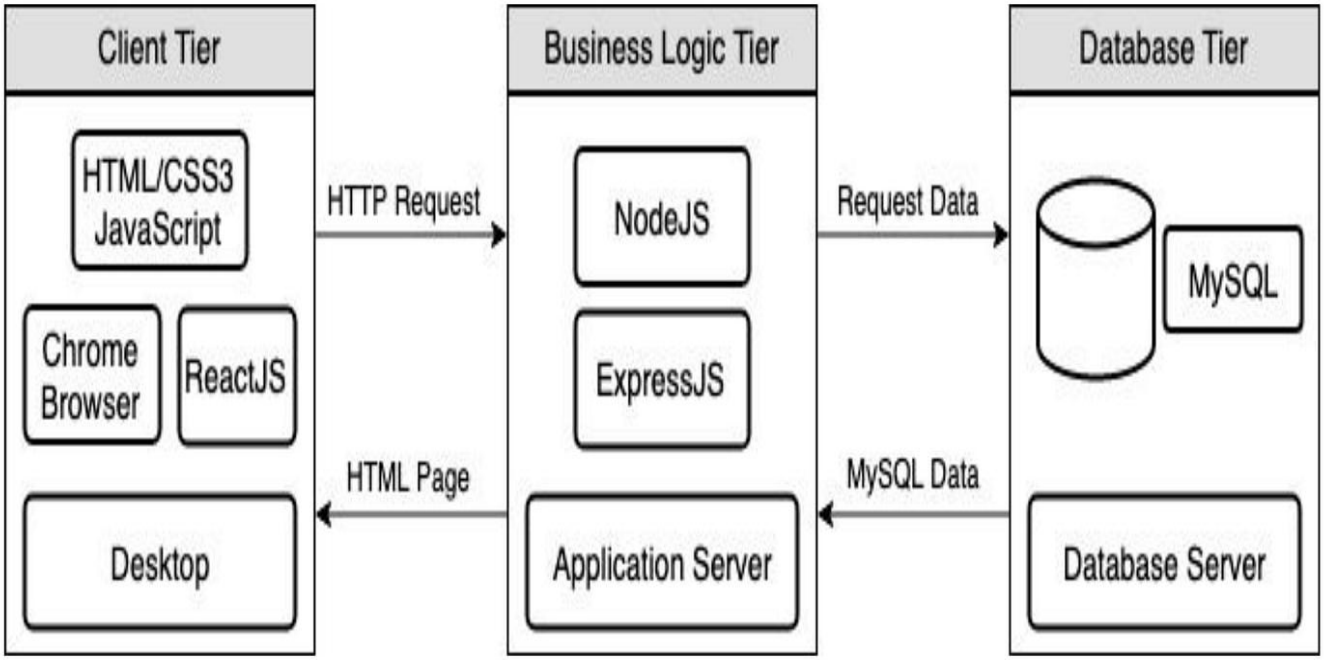
The design phase involved planning the structure of the web application. It included creating wireframes and UI layouts.

React.js was used for designing the frontend interface. Components were created for different sections of the application.

CSS and Bootstrap were used for styling and responsiveness.

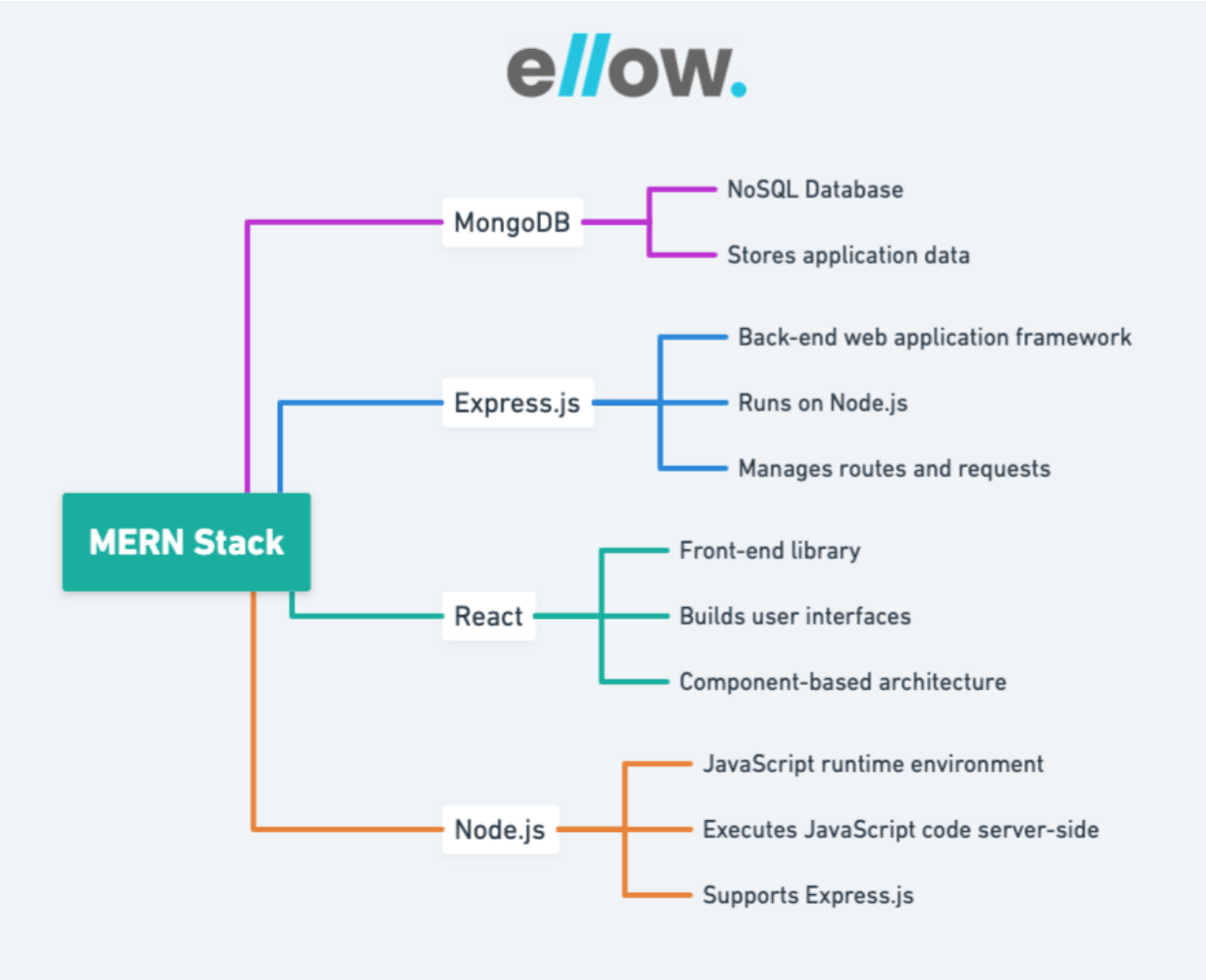
The backend design included creating APIs using Express.js. Node.js handled server-side logic.

MongoDB was used for database design and data storage.
Proper design ensured smooth functionality and user experience.



METHODOLOGIES USED

- The project followed an agile methodology. Development was done in small phases.
- Each phase included planning, development, testing, and deployment.
- Version control was maintained using Git.
- RESTful APIs were used for communication between frontend and backend.
- Debugging and testing were done regularly.
- This methodology ensured efficient and organized development



HARDWARE & SOFTWARE USED

- Hardware included a computer system with internet connectivity.
- Software tools included VS Code, Node.js, MongoDB, and browser tools.
- GitHub was used for version control.

Hardware Requirements

The development and execution of the MERN stack application required a system with adequate performance capabilities to handle multiple development tools simultaneously.

The primary hardware used was a personal computer or laptop with a minimum of **8 GB RAM**, which ensured smooth multitasking between code editors, browsers, and database services. A **64-bit processor (Intel i5 or higher / AMD equivalent)** was preferred to efficiently run Node.js and other development environments.

A **minimum of 256 GB storage (SSD recommended)** was used to ensure faster file access and improved system performance. SSD storage significantly reduced application loading time and improved development efficiency.

A stable **internet connection** was essential for downloading dependencies, accessing online documentation, and using cloud-based services like MongoDB Atlas.

Software Requirements
The software tools used during the internship were essential for developing, testing, and managing the MERN stack application.

1. Code Editor
The primary code editor used was Visual Studio Code.
It provides a lightweight yet powerful environment with features like syntax highlighting, debugging tools, and extensions for JavaScript, React, and Node.js development. Extensions such as Prettier and ESLint helped maintain clean and consistent code.
2. Runtime Environment
Node.js was used as the backend runtime environment.
It allows execution of JavaScript code on the server side and supports asynchronous programming, which improves performance and scalability of web applications.
3. Backend Framework
Express.js was used to build RESTful APIs.
It simplifies server-side development by providing routing, middleware support, and request handling mechanisms.

RESULTS

- The project was successfully completed with all required features.
- The application performed CRUD operations efficiently.
- User interface was responsive and user-friendly.
- Backend APIs worked smoothly.
- The project demonstrated full-stack development skills.
- By the end of the internship, I successfully developed a full-stack web application.
- I gained confidence in working with MERN stack technologies.
- I improved my coding efficiency and logical thinking.
- The project met all the required objectives.
- This internship enhanced my technical and professional skills.

FUTURE SCOPE

- The project can be enhanced with advanced features.
- Authentication systems can be improved.
- Cloud deployment can be implemented.
- Performance optimization can be done.
- More features like real-time updates can be added

The project can be improved by adding advanced features such as:

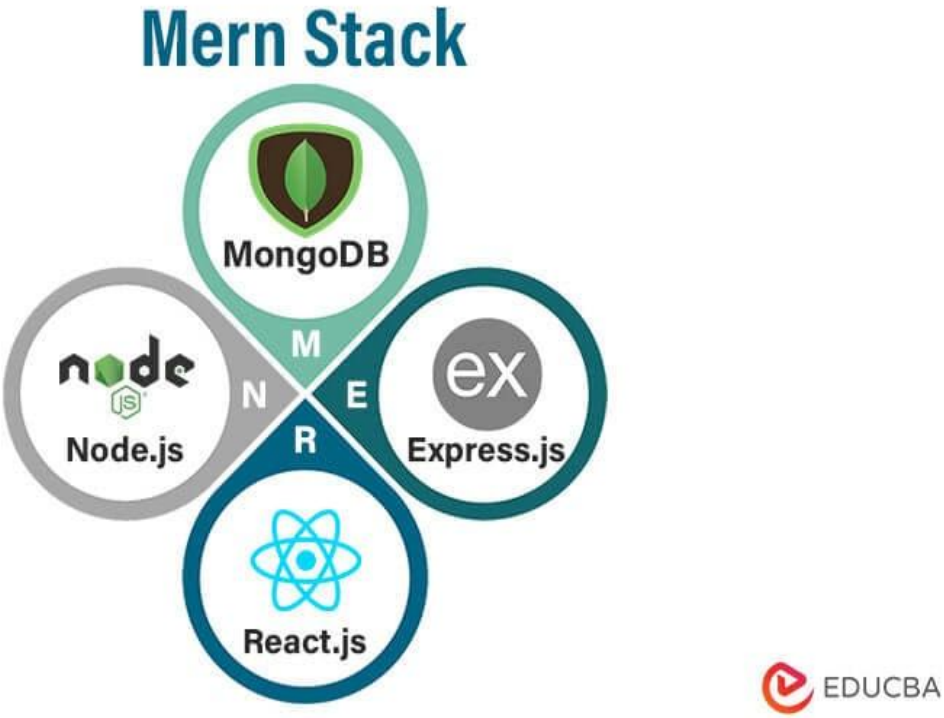
- User authentication with JWT
- Cloud database integration
- Deployment on platforms like Heroku
- Real-time updates using WebSockets
- Enhanced UI/UX design

  These improvements can make the application more scalable and efficient.

CONCLUSION

- The internship at CODEC Technologies was a great learning experience.
- It helped me understand MERN stack development in detail.
- I gained practical knowledge and improved my coding skills.
- The training prepared me for real-world challenges.
- Overall, it was a valuable step in my career.

The internship at CODEC Technologies provided me with valuable exposure to modern web development technologies.
It helped me bridge the gap between theoretical knowledge and practical implementation.
I gained hands-on experience in developing real-world applications.
The training improved my technical skills and confidence.
Overall, this internship has been a crucial step toward my career in software development.



REFERENCES

[1] React Documentation, Meta Platforms, Inc., 2023–2025. [Online]. Available: https://react.dev/

[2] Node.js Documentation, OpenJS Foundation, 2022–2025. [Online]. Available: https://nodejs.org/

[3] MongoDB Manual, MongoDB Inc., 2022–2025. [Online]. Available: https://www.mongodb.com/docs/

[4] Express.js Documentation, 2021–2024. [Online]. Available: https://expressjs.com/

[5] Brendan Eich, "JavaScript and Web Development Evolution," IEEE Computing Surveys, referenced 2022–2024.

[6] Ryan Dahl, "Node.js Architecture and Event-Driven Programming," IEEE/ACM Papers, 2021–2023.

[7] Jordan Walke, "React: Declarative UI Development," Meta Engineering Blog, 2022–

[8] "Modern Full Stack Development using MERN Stack," IEEE Xplore, 2022.

[9] "Scalable Web Applications using Node.js and MongoDB," IEEE Access Journal, 2023.

[10] "Frontend Optimization using React and Virtual DOM," International Journal of Computer Science, 2024.

[11] "Cloud-Based MERN Applications and Deployment Strategies," IEEE Conference Paper, 2023–2025.